

ONC201 Phase II Oncology Immunotherapy Trial

Edit Check Specification and CRF Completion Guidelines

Document purpose: Portfolio-ready clinical data management specification for a fictional longitudinal oncology study built in REDCap.

Scope: Screening through Safety Follow-up, including eligibility, demographics, registration, medical history, vital signs, laboratory results, tumor assessments, treatment, adverse events, concomitant medications, end of treatment, and follow-up.

Important: This is a synthetic educational portfolio document. It is not a sponsor-approved protocol, CRF completion guideline, medical standard, or production validation specification.

Study ID	ONC201
Study Type	Fictional Phase II oncology immunotherapy trial
Therapeutic Area	Melanoma / Immuno-oncology
EDC	REDCap (portfolio simulation)
Planned Events	Screening, Baseline, Week 4, Week 8, Week 12, Week 18, Week 24, End of Treatment, Safety Follow-up
Key Data Domains	Eligibility, safety, laboratory, tumor response, treatment exposure, concomitant medications, protocol deviations

1. CRF Design and Completion Guidelines

These guidelines define how the fictional ONC201 CRFs should be designed and completed consistently in REDCap.

General data entry

Enter data from the appropriate source record. Do not infer missing clinical values. Use the protocol-defined units and date format. Leave a value blank only when genuinely unavailable or not applicable through branching logic.

Record identifiers

Use a unique study record ID. Do not place participant names, initials, addresses, or other direct identifiers in the portfolio dataset.

Dates

Use a consistent Y-M-D date format. Chronology should be reviewed across consent, screening, eligibility, treatment, assessments, adverse events, and follow-up.

Required fields

Use required status for critical variables where appropriate, recognizing that standard REDCap data-entry forms may still allow a user to save after a warning.

Branching logic

Hide fields that are not applicable. Example: laboratory analytes display only when Sample Collected = Yes; pregnancy-test fields display only when applicable; seriousness details display only when Serious AE = Yes.

Calculated fields

Do not manually enter calculated values. Save source fields first, then verify calculations such as BMI, age at screening, and tumor percent change.

Repeating instruments

Use repeating instances for domains such as Adverse Events, Concomitant Medications, Medical History, and Protocol Deviations when multiple records can occur.

Corrections

In a real production EDC, corrections should preserve audit trail and reason-for-change requirements. For this portfolio, document simulated corrections in the Query Tracker.

Missing data

Distinguish expected blanks caused by branching logic from true missing data. A hidden non-applicable field should not be queried as missing.

Clinical significance

A value outside a reference range is not automatically a data error. Query implausible, inconsistent, missing, or protocol-critical data; do not replace clinical judgment with a range check.

2. Instrument-by-Instrument CRF Guidelines

Screening

Capture consent status/date, screening date, screening number and key screening assessments. Consent must precede study-specific procedures. If Consent Obtained = No, downstream consent-dependent fields should remain hidden or incomplete.

Subject Registration

Capture study ID/record ID and Date of Birth if included in the fictional dataset. DOB is a text box with date validation, not a calculated field.

Demographics

Capture screening date, sex at birth and demographic variables. Age at Screening should be calculated from DOB and Screening Date; in a longitudinal project use event-qualified references when the source fields are stored in different events.

Eligibility Confirmation

Document all inclusion/exclusion criteria and final eligibility. Randomization/treatment data should only be entered for participants who meet final eligibility requirements.

Medical History

Enter one clinically relevant condition per repeating instance when the instrument is configured to repeat. Include start date/year, ongoing status, and end date when applicable.

Vital Signs

Capture assessment date/time where applicable, height, weight, blood pressure, heart rate and temperature. BMI should calculate from height and weight; do not type the BMI manually.

Laboratory Results

First record whether a sample was collected. If Sample Collected = No, analyte fields such as WBC and ANC should be hidden. Record units consistently. Clinically Significant should reflect investigator/site assessment, not merely an automated range flag.

Tumor Assessment

Capture assessment date, target lesion sum, non-target status, new lesions, overall response and progression. Baseline should establish the reference tumor burden; follow-up visits may calculate percent change from baseline. Baseline response itself should not be labeled as CR/PR/SD/PD unless the study design explicitly calls for it.

Treatment Administration

Capture treatment date, whether treatment was given, study drug, planned/actual dose, modifications, delays, completion and infusion reaction. Dose modification reason should appear only when Dose Modified = Yes.

Adverse Events

Use one repeating instance per AE. Capture term, onset/end dates, grade/severity, seriousness, relationship, action, outcome and seriousness details when applicable. End date cannot precede onset date.

Concomitant Medications

Use one repeating instance per medication. Capture medication name, indication, start/end dates, dose/unit/frequency and ongoing status. Ongoing medications should not require an end date.

Protocol Deviations

Capture deviation date, category, description, severity, root cause, corrective action, resolution status and resolution date. Resolution date should be required only when resolved.

End of Treatment

Capture last treatment date, treatment completion/discontinuation, reason for discontinuation, disease status and required end-of-treatment assessments.

Safety Follow-up

Capture follow-up date, ongoing/resolved AEs, subsequent safety information and survival/status fields defined for the portfolio.

3. Edit Check Specification

The checks below are portfolio examples. Exact variable names and event unique names should be aligned to the final REDCap data dictionary.

Check ID	Form	Field	Rule	Priority	Suggested Query
ONC-EC001	Screening	Consent	Consent must be Yes before study-specific screening procedures.	Critical	Consent not documented before screening. Please verify consent status/date.
ONC-EC002	Screening	Consent Date	Consent date cannot be after screening date.	Critical	Consent date is after screening date. Please verify both dates.
ONC-EC003	Demographics / Registration	Age at Screening	Age at screening must calculate from DOB and screening date.	High	Calculated age is missing or inconsistent with DOB/screening date. Please verify source dates.
ONC-EC004	Eligibility	Final Eligibility	Randomization/treatment must not occur when final eligibility = No.	Critical	Participant is not eligible but randomization/treatment data are present. Please review.
ONC-EC005	Eligibility	Criteria completeness	All protocol-critical inclusion/exclusion criteria must be completed before eligibility confirmation.	High	Eligibility is confirmed but one or more criteria are incomplete.
ONC-EC006	Randomization	Randomization Date	Randomization date must be on/after eligibility confirmation and consent.	Critical	Randomization chronology is inconsistent. Please verify dates.
ONC-EC007	Vital Signs	Height	Height must be present when BMI is expected.	Medium	BMI is expected but height is missing.
ONC-EC008	Vital Signs	Weight	Weight must be present when BMI is expected.	Medium	BMI is expected but weight is missing.
ONC-EC009	Vital Signs	BMI	BMI must equal weight(kg) / height(m)^2 within rounding tolerance.	Medium	Calculated BMI is inconsistent with recorded height/weight.
ONC-EC010	Laboratory	Sample Collected	If Sample Collected = No, laboratory analytes should be blank/hidden.	High	Laboratory result is present although Sample Collected = No.
ONC-EC011	Laboratory	ANC	ANC should be present when the required laboratory sample is collected, unless documented unavailable.	High	ANC is missing for a collected required laboratory sample.
ONC-EC012	Laboratory	WBC	WBC should be present when the required laboratory sample is collected, unless documented unavailable.	High	WBC is missing for a collected required laboratory sample.
ONC-EC013	Laboratory	Creatinine	Creatinine should be numeric and use the specified unit.	Medium	Creatinine is missing, nonnumeric, or unit is inconsistent.
ONC-EC014	Laboratory	Bilirubin	Total bilirubin should be completed for required visits.	Medium	Total bilirubin is missing at a required laboratory visit.
ONC-EC015	Laboratory	Clinical Significance	Clinically Significant should be completed when laboratory review is required.	Medium	Clinical significance assessment is missing.
ONC-EC016	Tumor Assessment	Assessment Date	Post-baseline tumor assessment date must occur after baseline.	High	Tumor assessment chronology is inconsistent.
ONC-EC017	Tumor Assessment	Target Sum	Target lesion sum should be numeric and nonnegative.	High	Target lesion sum is missing or invalid.
ONC-EC018	Tumor Assessment	Percent Change	Percent change should equal ((current target sum - baseline target sum) / baseline target sum) x 100.	High	Percent change does not match baseline/current target lesion sums.
ONC-EC019	Tumor Assessment	New Lesions	If New Lesions = Yes, overall response/progression should be reviewed for consistency.	High	New lesions are reported but response/progression fields may be inconsistent.
ONC-EC020	Tumor Assessment	Progression Date	If Disease Progression = Yes, progression date should be completed.	High	Disease progression is Yes but progression date is missing.
ONC-EC021	Tumor Assessment	Response	Progressive Disease should be reviewed when progression is marked No.	High	Overall response and progression flag are inconsistent.
ONC-EC022	Treatment	Treatment Given	If Treatment Given = No, actual dose should be blank or zero per convention.	High	Actual dose is present although treatment was not given.
ONC-EC023	Treatment	Actual Dose	If Treatment Given = Yes, actual dose should be completed.	High	Treatment was given but actual dose is missing.
ONC-EC024	Treatment	Dose Modified	If Dose Modified = Yes, modification reason must be completed.	Medium	Dose modification is Yes but reason is missing.
ONC-EC025	Treatment	Treatment Delay	If Treatment Delay = Yes, delay reason must be completed.	Medium	Treatment delay is Yes but reason is missing.
ONC-EC026	Treatment	Treatment Date	Treatment date must not precede consent/randomization as applicable.	Critical	Treatment date precedes required study milestone. Please verify.
ONC-EC027	Treatment	Infusion Reaction	If infusion reaction is observed, review for corresponding AE entry when required.	High	Infusion reaction is reported without a corresponding AE record/review.
ONC-EC028	Adverse Events	AE Start/End Date	AE end date must be on/after AE start date.	High	AE end date precedes AE start date.
ONC-EC029	Adverse Events	Serious AE	If Serious AE = Yes, seriousness details/criterion must be completed.	Critical	AE is serious but seriousness details are missing.
ONC-EC030	Adverse Events	AE Grade	AE grade/severity must be completed for every AE.	High	AE grade is missing.
ONC-EC031	Adverse Events	Outcome	Resolved/completed AE should have an outcome and end date as applicable.	Medium	AE outcome/end-date information is incomplete.
ONC-EC032	Adverse Events	Relationship	Relationship to study treatment should be completed.	High	AE relationship assessment is missing.

Check ID	Form	Field	Rule	Priority	Suggested Query
ONC-EC033	Concomitant Medications	Medication Dates	Medication end date cannot precede start date.	High	Concomitant medication end date precedes start date.
ONC-EC034	Concomitant Medications	Ongoing	If Ongoing = Yes, end date should normally be blank.	Medium	Medication is marked ongoing but an end date is entered.
ONC-EC035	Concomitant Medications	Indication	Medication indication should be completed.	Low	Concomitant medication indication is missing.
ONC-EC036	Protocol Deviations	Resolution	If Resolved = Yes, resolution date should be completed.	Medium	Deviation is resolved but resolution date is missing.
ONC-EC037	Protocol Deviations	Severity	Deviation severity/category should be completed.	Medium	Protocol deviation classification is incomplete.
ONC-EC038	Visit Review	Visit Sequence	Scheduled visit dates should follow chronological order.	High	Visit chronology is inconsistent with the planned sequence.
ONC-EC039	Visit Review	Required Assessments	Required CRFs/assessments should be present at each applicable event.	High	One or more required assessments are missing for this event.
ONC-EC040	End of Treatment	Discontinuation Reason	If treatment is discontinued early, reason should be completed.	High	Early discontinuation is documented without a reason.
ONC-EC041	End of Treatment	Last Treatment Date	Last treatment date should not precede first treatment date.	High	Last treatment date is inconsistent with treatment history.
ONC-EC042	Safety Follow-up	Follow-up Date	Safety follow-up should occur after End of Treatment.	High	Safety follow-up date precedes End of Treatment.
ONC-EC043	Safety Follow-up	Ongoing AEs	AEs ongoing at EOT should be reviewed at safety follow-up.	High	An AE was ongoing at EOT but follow-up status is missing.
ONC-EC044	Cross-form	Sex / Pregnancy Test	Pregnancy-test fields should appear only when applicable according to study logic.	Medium	Pregnancy-test data are inconsistent with applicability logic.
ONC-EC045	Cross-form	Record ID	Every repeating/visit record must belong to a valid participant.	Critical	Orphan clinical record detected. Please verify participant ID.

4. REDCap Logic Examples

Use the exact variable names and event unique names from the final project. In longitudinal REDCap projects, event-qualified references follow the pattern [event_unique_name][variable].

Laboratory branching

Show analytes only when sample collected: [sample_collected] = '1'

Serious AE branching

Show seriousness details only when Serious AE = Yes: [ae_serious] = '1'

Dose modification branching

Show dose modification reason only when Dose Modified = Yes: [dose_modified] = '1'

Age at screening

If DOB and Screening Date are in the same event: `datediff([dob],[screening_date],"y","ymd",true)`. If they are in different events, qualify each field with its event unique name.

Tumor percentage change

$((\text{current target lesion sum} - \text{baseline target lesion sum}) / \text{baseline target lesion sum}) \times 100$. Example: baseline 50 and current 40 gives -20%. The calculated field stores the numeric result.

5. Data Review and Query Workflow

1. Run built-in or custom REDCap Data Quality rules when the module is available.
2. Review critical variables first: consent, eligibility, treatment, safety, laboratory, tumor response, and key dates.
3. Review listings/reports for cross-form checks that are difficult to implement as simple rules.
4. Document each discrepancy in the Query Tracker with subject, event, form, field, issue type, priority, query text, status, response, and resolution.
5. Do not change source-derived data merely to make a check pass. Simulated corrections in this portfolio should be documented as if supported by a site response.
6. Re-run the applicable check after correction and close the query only when the discrepancy is resolved.
7. Before simulated database lock, review open critical/high-priority queries, missing critical data, unresolved SAEs, ongoing AEs, protocol deviations, and calculated fields.

6. Portfolio / GitHub Use

Recommended filename: **ONC201_Edit_Checks_and_CRF_Guidelines.pdf**

Place this document under a folder such as **Documentation/** or **Clinical_Data_Management/**. In the README, describe it as a synthetic CRF completion and edit-check specification created to demonstrate REDCap database design, clinical data review, query management, and oncology data-quality concepts.

Do not represent this document as an approved sponsor specification or as guidance used on a real clinical trial.

Reference note

REDCap best-practice resources emphasize defining critical variables, using field notes and realistic range checks, applying branching logic to hide non-applicable fields, and using custom data-quality rules/listings to identify missing or inconsistent data. Longitudinal field references may require event-qualified notation.